MLT LED SERIES
Service,
Maintenance and
Operation Manual

- MLT1920-LED
- MLT2560-LED
- MLS2560-LED
- MLS3200-LED
- MLT3840-LED
- MLS3840-LED

Contents

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Complete this section on your new Lighting Tower:
Owner Details
Owner:
Owners Representative:
Phone No:
Address:
Email:

Equipment Details
Delivery Date:
Machine Model:
Machine Serial No:

As Per Compliance Documents

Call Sign No:

As Per Compliance Documents

Engine Type:

As Per Compliance Documents

Engine Serial No:

As Per Compliance Documents

Generator Type:

As Per Compliance Documents

Generator Serial No:

As Per Compliance Documents

Controller Type:

As Per Compliance Documents

Controller Serial No:

As Per Compliance Documents

GPS Serial No:

As Per Compliance Documents

Light Type and Qty:

As Per Compliance Documents

In order to receive maximum benefit from your Mickala product, please read this manual carefully and follow the operating instructions.
All information (including illustrations and pictures) contained in this manual are correct at the time of publication. Mickala Lighting Towers reserves the right to make changes in this manual at any time without prior notice. Mickala Mining Maintenance takes no

responsibility for the
accuracy or otherwise of the information contained within this document.

Company Summary
Mickala Lighting Towers business philosophy highlights the importance of forming close
alliances and long term relationships with our clients. At the core of this philosophy is a
complete understanding that a sustainable long-term business relationship can only be
sustained with a strong commitment to providing a high quality service.
In such a highly competitive industry, Mickala Mining Maintenance values its personal touch
and combined with our attention to safety, quality, environment and continuous improvement
this sets Mickala Mining Maintenance apart from its competitors.
Mickala Lighting Towers has a team dedicated to providing exceptional service to ensure that
experiences with Mickala Mining Maintenance are an enjoyable and unforgettable one. We
pride ourselves on exceptional service to all of our clients and we will go above and beyond to
ensure client satisfaction. Our services will be provided in a safe, ethical and highly
professional manner.
At Mickala Lighting Towers we strive to continually provide the highest standard of service to
our clients through process management.
In doing so we aim to:
Ensure that our safety performance is of the highest standard for both our clients and
our people
Assist our clients to achieve their own goals
Respond to our client’s needs promptly and appropriately
Anticipate our clients requirements ahead of time and ahead of our competitors

Our Contact Details
Sales and Enquiries
A: QLD: 6 Turbo Drive, PAGET QLD 4740
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Business Unit
Document Title
Operations
Mickala Lighting Tower LED Series Operation Manual
Author
Approved by
Date Approved
Revision Due Date
Maintenance Manager

Introduction and Product Details

Safety

Please familiarise yourself with the safety precautions and the Lighting Tower operation procedures contained within this manual prior to using the equipment. Failure to do so may cause operational problems, which could affect your warranty and/or result in serious injury.
Mickala Mining Maintenance can conduct safety training for personnel who are required to operate this equipment. Please contact us if you require company specific training.

General

For many years, Mickala Lighting Towers has been perfecting the art of Lighting Tower solutions. We have the ability to design and manufacturer our own reliable Lighting Towers that we know truly benefits the mining industry. We guarantee our Lighting Towers are advanced in technology and that they are capable of handling any situation.
Mickala Lighting Towers is very serious about what we do. We are that serious it has taken many years of experience in the mining industry to allow us to design and engineer our own Mickala Lighting Towers. Our Lighting Towers are more efficient, easily maintained, robust and cost-effective while offering the highest environmental sensitivity possible.
Our Lighting Towers are engineered and certified to suit Australian Mining conditions.
Mickala's Lighting Towers comply with the following Australian and Mining standards:
Mines Safety and Inspection Regulations 1995
AS2790 Transportable Generator Sets
AS3713 Noise Level Emissions
AS3001 Electrical Wiring Rules
AS3010.1 Supply of Generator Set
AS1170.2 Wind Loading Criteria
MDG 41
MDG 15
This Operation Manual has been prepared to assist in the operation of your Mickala Lighting Tower. Using this manual in conjunction with the Parts Manual (available separately) will help to ensure that the Lighting Tower operates at maximum performance and efficiency for many years.
Please note that in dirty or dusty environments more attention must be paid to frequent servicing to keep the Lighting Tower running efficiently.

Spare Parts and Support

In all correspondence with Mickala Lighting Towers, always have the machine serial number available to allow our company to assist you promptly.
Always ensure that adjustments and repairs are done by personnel who are authorised to do the work and have been properly trained. Carrying out unauthorised works or modifying your Lighting Tower without Mickala’s authority can void your warranty. Mickala Lighting

Towers is
able to supply all parts, labour and service items required for any of our Lighting Towers.

Components of Lighting Tower
MLT3840-LED model displayed for illustration purposes

Light Head
Assembly

Light Head
Rotating Assembly

Mast Assembly

Engine / Generator
Module

Spare Tyre
(optional)

Control Panel
(Electrical / Hydraulic)

Hydraulic Outrigger
Assembly
Jockey Wheel

3t Tow Hitch

Standard Inclusions

The MLT LED Series include the following features as standard inclusions:

Pressure Relief Radiator Cap
Starter and Battery Isolator
Jump Start Receptacle
E Stop
Auto Start/Stop
Low Level Fuel Beacon
Safety Valves on all Hydraulic Rams
Wheels Chocks
Durable bump and wear packages for sled mounted towers
Amber Flashing Light
Red Low Fuel Beacon
Fail Safe Shutdown System

- Engine Temperature
- Coolant Temperature
- Oil Level

LED Tail Lights
4.5kg Fire Extinguisher
LED Work Lights
MDG15/41 Compliant
3 Stage Electronic Paint Protection

Operating Instructions
Control Panel Identification

Lighting Tower
Controller

Outrigger Operations

Boom Operations

Pre Start Checks

1 - Check filter service is up to date (500 hour service intervals). If changing ,write last service hours on filter

2 - Check engine oil level

3 - Check radiator coolant

4 - Check hydraulics fluid level

5 - Check machine for damage

6 - Check for leaks

7 - Visually inspect electrical conduits for damage

8 - Ensure lenses are clean

9 - Check operation of all machine functions

NOTE: Any issues found in steps 1 – 8 must be rectified by a qualified tradesperson before performing step 9. If the Lighting Tower fails any checks, follow site specific procedures for tagging the machine “OUT OF SERVICE” and call for a service

Lighting Tower Operation Procedure
1. Lower all 4 outriggers until level site glass bubble is centred
Note – Do not forget to remove the travel strap from mast

2. Raise mast until it is in a vertical position. For full scope extension, boom must be vertical

3. Raise telescopic ram until fully raised (or preferred height)

4. Tilt light head until in a vertical position

5. Rotate light head until desired position is reached

6. Commence Start Up procedure as detailed (pg16)
MLT2560-LED model displayed below for illustration purposes

Start Up Procedure
WARNING
Authorised personnel only to operate
Ensure that the Lighting Tower is positioned on level ground at all times and is fundamentally stable
Ensure all personnel are clear of the engine when starting
Be aware of overhead power lines
Do not operate lighting tower unless the daily pre start has been carried out

To Start Engine:
Note: Ensure both Battery and Starter Isolators are turned on
1.
2.
3.

Press Manual Start button
Press green Start button
Press Light On button to turn lights on.
Continue to press button until all lights are activated

3

1

2

To Start With Auto Timer Mode:
Note: Ensure both Battery and Starter Isolators are turned on
1.
2.
3.
4.
5.

Press Auto button
Select 01 – Auto Timer Mode
(using “▫” and “▫” buttons)
Press Auto button
Select 01 – Timer Start
Press Auto button

2

4

1,3,5

Machine will start in Auto Timer Mode

Shutdown Procedure

To Stop Engine and Shutdown Lights:

1. Press Stop button

1

Hydraulic Schematics

Hydraulic Schematics

Engine Specifications

Servicing
Mickala Lighting Towers requires each Lighting Tower to be serviced every 500 hours. The 500 Hour Service Report outlines the necessary inspection requirements to maximise the life of the Lighting Tower.
Mickala Lighting Towers requires a copy of the completed 500 Hour Service Report (up to the earlier of either - 12 months from the date of delivery or 1500 hours of operation) as outlined in the warranty conditions.
If you require additional copies of the Service Report please contact us on (07) 4998 5447.

Filter Locations
1.
2.
3.
4.
5.

External Oil Filter
Fuel Filter
Air Filter
Hydraulic Return Filter (Changed every 2000hrs)
Oil Filter

2. External Oil Filter

1. Fuel Filter

3. Air Filter

4. Hydraulic Return Filter

5. Oil Filter

Electrical
12V Inputs From Controller
24V From Generator

Green light on = System ok
Red Light = System Fault
(1 Per Circuit)

Glow Plug

Hold Circuit

Start/Stop Circuit

Low Coolant Module

Proximity Limit Switch Relays

Ignition Relay

24 Volt Main From Controller
12V Main Breaker

Engine Tune Specifications MLT3200-LED Skid (RFL Generator)
Engine RPM Unloaded = 2480
Engine RPM Loaded = 2260
Generator Voltage Unloaded = 34.6
Generator Voltage Loaded = 26.1

Engine Tune Specifications MLT2560-LED Dual Axle Trailer (RFL Generator)
Engine RPM Unloaded = 2450
Engine RPM Loaded = 2260
Generator Voltage Unloaded = 34.3
Generator Voltage Loaded = 27.6

Shut Down Parameters

Engine RPM Adjustment

Engine RPM
Adjustment Bolt

Hydraulic Relief Pressure

Hydraulic Pressure test Port

Hydraulic Relief Adjustment
180-190 Bar

Grease Points

Cross over relief pressure

Hydraulic Gearbox Grease Points
Caution - Excessive Grease will pop front seal out

Grease Points

Gearbox Grease Relief

Wheel Bearing

Safety
This section is for information purposes only and is not a safety manual. Mickala Lighting Towers encourages thorough training and maintenance of high safety standards in the use of this equipment. Responsibility for compiling proper safety instructions rests with the owner of the product. For any further information, contact
Mickala Mining Maintenance.

General
The Lighting Tower is designed to be safe when used in the correct manner. Responsibility for
safety however, rests with the personnel who install, use and maintain the Lighting Tower. The
following safety precautions, if followed, will minimise the possibility of accidents. Before
performing any procedure or operating technique, it is up to the user to ensure that it is safe.
The Lighting Tower should only be operated by personnel who are authorised and trained.
Read and understand all safety precautions and warnings before
operating or performing maintenance on the Lighting Tower
Failure to follow the instructions, procedures, and safety precautions in
this manual may increase the possibility of accidents and injuries
Never start the Lighting Tower unless it is safe to do so
Install and operate this Lighting Tower only in full compliance with
relevant National, Local, or Federal Codes, Standards or other
requirements

Handling and Towing
This manual should be read before moving/lifting the Lighting Tower, or towing a mobile
Lighting Tower.

Lifting
✓
✓
✓
✓
✓

Do not lift Lighting Tower with a forklift
Only qualified personnel are to use lifting equipment
Do not lift the Lighting Tower by the mast. (Use ALL lifting points provided)
Ensure the lifting equipment is in good condition and has a capacity suitable for the load
Keep all personnel away from Lighting Tower when it is suspended. Use a tag line from the
pintle hook to assist in stabilising the equipment

Securing Positions
✓ Lighting Tower should only be secured to a carrying vehicle by restraining around tyres and
tie down lugs fitted to the underside of the Lighting Tower chassis
✓ Do not chain or sling across the drawbar or mast as damage may occur
Ensure the lifting, rigging and supporting structure is in good condition
and has a capacity suitable for the load
Keep all personnel away from the Lighting Tower when it is suspended.
DO NOT STAND UNDER SUSPENDED LOAD

Before Towing
√ Take care not to reverse vehicle into Lighting Tower, as this may damage brake components. Use a Spotter at all times
√ Check vehicle towing hitch load rating and ball/hitch size and type are compatible with
Lighting Tower (this can be found on the compliance/manufacture plate)
√ Check mast is in the mast support, that it is fully retracted and that the transport strap is
secured
√ Check doors are latched
√ Check tyres are correctly inflated and that the tread is roadworthy. Check tyre placard or
tyre sidewalls for recommended cold inflation pressures
√ Check all stabiliser legs are raised
√ Connect hitch, towing chains and electrical plug (if fitted) to vehicle
√ Check jockey wheel is fully retracted and secured horizontally
√ Store wheel chocks (if supplied)
√ Check that the brake reversing lock tab on hitch is open (if brakes are fitted)
√ Check brake operation, cables are not frayed, and that adjustment is correct
√ Check tail light operation
When towing a mobile Lighting Tower, observe all Codes, Standards or
other regulations and traffic laws. These include those regulations
specifying required equipment and maximum and minimum speeds
Maximum recommended towing speed on sealed surfaces is 80 km/h
(dependent on conditions and local limits)
Maximum recommended towing speed on unsealed surfaces is 40km/h
(dependent on conditions, local limits and site conditions)
Maximum recommended towing speed for skid mounted plant is 15km/h
Do not permit personnel to ride in or on the Lighting Tower. Do not
permit personnel to stand or ride on the drawbar or to stand or walk
between the Lighting Tower and the towing vehicle.

Unhitching Lighting Tower
√ Check that the ground is level, the surface secure, and the position is not too close to a
dropping embankment
√ Check parking brake is applied on the towing vehicle, and apply parking brake on Lighting
Tower
√ Install wheel chocks (if supplied) ensuring that the Lighting Tower will not roll down any
incline - park machine in a fundamentally stable location
√ Remove chains and electrical plug (if fitted)
√ Set the jockey wheel in place ensuring that the swivel plate locks into the vertical position
√ Raise Lighting Tower from hitch using the jockey wheel
√ Lower stabiliser legs
Ensure a Pre-Start checklist is completed before each
Requirements will vary based on the location of application.

Set Up
To receive optimum Lighting Tower performance set the Lighting Tower 40 to 60 metres from the work area.

Before Starting Engine
✓ Check all conduits for splits, bare wires or unsecured conduit fittings
✓ Check that all circuit breaker boards are secure and sealed
✓ Check that all covers and caps are secure on alternator, cooling fan/radiator, engine and fuel tank
✓ Check Lighting Tower is level on secure ground and stabiliser legs are down, with wheel chocks in place (if fitted)

Operating
✓ Before lowering mast, be sure inner mast is fully retracted
Do not operate within a confined space
Do not move Lighting Tower with generator running, or before the mast is retracted in its cradle
Ensure operating area is clear of overhead power lines and any other obstruction before raising mast
Do not operate with a damaged or missing lens or an open light, otherwise skin/eye damage can occur
Avoid looking directly at lights or standing within 8 metres of lights when operating
Do not attempt to operate the Lighting Tower with a known unsafe condition. If the Lighting Tower is unsafe, tag the machine out of service (per site procedures) and isolate the battery (using battery isolator) so that it cannot be started until the condition is corrected
Do not operate the Lighting Tower in any classification of hazardous environment
Do not operate the Lighting Tower if there is risk of wind gusts greater than 100 km/h

Hazards

Fire and Explosion

Fuels and fumes associated with Lighting Towers can be flammable and potentially explosive.

Proper care in handling these materials can dramatically limit the risk of fire or explosion.

However, safety dictates that fully charged fire extinguishers are kept on hand. Personnel must know how to operate them.

- Never store flammable liquids near the engine
- Do not leave rags inside the engine compartment
- Do not smoke or allow sparks, flames or other sources of ignition around fuel or batteries
- Fuel vapours are explosive. Hydrogen gas generated by charging batteries is also explosive
- Turn off or disconnect the power to the battery charger before making or breaking connections with the battery
- Keep grounded conductive objects, such as tools, away from exposed live electrical parts, such as terminals, to avoid arcing. Sparks and arcing might ignite fuel or vapours
- Do not refill the fuel tank while the engine is running
- Do not attempt to operate the Lighting Tower with any known leaks in the fuel system

Mechanical

The Lighting Tower is designed with guards for protection from moving parts. Care must still be taken to protect personnel and equipment from other mechanical hazards when working around the Lighting Tower.

- Do not attempt to operate the Lighting Tower with any safety guards or panels removed
- During operation, do not attempt to reach under or around the guards for any reason
- Ensure that the engine is operated only from the control panel or from the operators position by one person only
- Keep hands, arms, long hair, loose clothing and jewellery away from pulleys, belts and other moving parts
- Caution - Some moving parts cannot be seen clearly when the set is running

1.3

Keep access doors on enclosures, if equipped, closed and locked when not required to be open
Avoid contact with hot oil, hot coolant, hot exhaust gases, hot surfaces and sharp edges and corners
If your skin comes into contact with high pressure fuel, obtain medical assistance immediately
If your skin comes into contact with high pressure hydraulic oil, obtain medical assistance immediately
Wear protective clothing including gloves and hat when working around the Lighting Tower
Do not remove the radiator filler cap until the coolant has cooled. Then loosen the cap slowly to relieve any excess pressure before removing the cap completely

Chemical

Fuels, oils, coolants, lubricants and battery electrolyte used in this Lighting Tower are typical of the industry. However, they can be hazardous to personnel if not treated properly.
Do not swallow or have skin contact with fuel, oil, coolant lubricants or battery electrolyte. If swallowed, seek medical treatment immediately
Do not induce vomiting if fuel is swallowed. For skin contact, wash with soap and water
Do not wear clothing that has been contaminated by fuel or lube oil

Noise

Lighting Towers that are not equipped with sound attenuating enclosures can produce noise levels in excess of 85 dBA. Prolonged exposure to noise levels above 85 dBA is hazardous to hearing.
Ear protection should be worn when operating or working around the Lighting Tower

Electrical

Do not attempt to connect or disconnect load while standing in water or on wet or unstable ground
Be sure all electrical power is isolated from electrical equipment being serviced
Keep all electrical equipment clean and dry

Replace any wiring where the insulation is cracked, cut, frayed or otherwise degraded
Replace terminals that are worn, discoloured or corroded. Keep terminals clean and tight
Use only fire extinguishers suitable for electrical fires

First Aid for Electrical Injuries
This section is included for information purposes only and is not a First Aid manual. Medical advice should always be sought in the preparation of procedures and as soon as possible in an emergency.
While Mickala Mining Maintenance encourages thorough training of First Aid personnel, responsibility for compiling adequate First Aid instructions rests with the owner of the product.
For any further information, contact Mickala Mining Maintenance.
Shock
Most injured people show the signs and symptoms of shock. These are:

√ Pale, cool, moist skin
√ A weak, rapid pulse
√ Altered conscious states
√ Rapid breathing
√ Nausea / vomiting
√ Restlessness / irritability
√ Extreme thirst

Do not give any food, fluids or stimulants. Where possible, the conscious victim should be assisted to lie down in the most comfortable position with all injured parts supported. If injuries permit, raise both legs to boost the circulation to the heart and brain. The unconscious victim should be placed in the recovery position. Maintain the victim's body temperature.
Burns
Electrical burns are often deep, and the victim will have both an entrance and exit wound.
Although these wounds may look superficial, the tissues below may be severely damaged.
After ensuring the scene is safe:

√ Cool burns by flushing with cool water
√ Remove victims rings and jewellery (metal retains heat)
√ Cover the burn with a dry, sterile dressing
√ Take steps to minimise shock

In all emergencies call Emergency Services as soon as possible

Electrical Shock Fact Sheet

DRABCD Fact Sheet

Fault Diagnosis
In the event of experiencing difficulty in operating your Mickala Lighting Tower, refer to the table below. If the difficulty persists or is not a listed item, contact Mickala Mining Maintenance on (07) 4998 5447 for further support information.

Symptom

Possible Solution

Engine not starting

•
•
•
•
•
•

Engine shutting down

•
•
•
•
•

Lights Strobing/Blinking

• Low Voltage
• High Voltage

Circuit breakers tripping

• Turn engine off. Visually check for damaged cables. If no damage apparent, reset circuit breakers. Follow start up procedure
If fault persists after resetting, do not operate tower and call for service

Lights fail to illuminate

• Check for damage to lighting circuitry
• Check All light indicators are Green, If circuit is red call for Service

Hydraulic pump and motor does not operate

• Engine not running
• Check battery isolator is in ON position
• Check main hydraulic fuse

Check both isolators are in the ON position
Check Emergency Stops are not engaged
Check battery is charged. Charge or replace as required
Check sufficient fuel in tank
Check fuel tank for water contamination

If tank has previously been emptied, the fuel system may
require priming

Check control panel for any faults
Engine Over/Under speed
High Temp
Low Level Coolant
Check engine fluid levels
Engine and alternator will be very hot after
running. Allow to cool before checking fluid levels
• Check sufficient fuel in tank

Ta

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